

The GUE spectrum as a determinantal point process - in particular, in terms of the correlation kernel K_N (see [this collection of blog posts](#) by Tao for more.)

Refer to [1, Chapter 2.6.4] for the bulk of the argument, but also [2, Chapter 5.5] and [3] if accessible.

1. **Deriving the kernel K_N .**
 2. **Properties of K_N and obtaining the expected empirical distribution.**
 3. **(Future) Continued analysis of K_N to derive Wigner's semicircle law.**
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The joint probability distribution for the spectrum of a Gaussian Unitary Ensemble (GUE) is

$$\rho_N(\lambda) = C_N e^{-|\lambda|^2/2} |\Delta_N(\lambda)|^2, \quad (1)$$

up to a constant C_N , as given in [1, 3], for instance. Here $\lambda = (\lambda_1, \dots, \lambda_N)$ and Δ_N is the Vandermonde determinant

$$\Delta_N(\lambda) = \prod_{1 \leq i < j \leq N} (\lambda_i - \lambda_j). \quad (2)$$

Interpret $|\lambda|^2$ as the Euclidean norm of λ . The goal here is to understand (1) as a determinantal form.

1. Deriving the kernel K_N . The product in (2) is called the Vandermonde determinant because, up to sign, it can be expressed as

$$\Delta_N(\lambda) = \det \begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{N-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_N & \lambda_N^2 & \cdots & \lambda_N^{N-1} \end{bmatrix} =: \det(V(\lambda)),$$

i.e. the determinant of the Vandermonde matrix V . One way to show this is to use the fact that applying row or column operations to a matrix does not alter its determinant: this is used several times henceforth.

With basic properties of the determinant we have $|\Delta_N(\lambda)|^2 = \det(VV^T)$. Furthermore, by explicitly writing out V and V^T , or by considering $V = [\lambda_i^{j-1}]_{1 \leq i, j \leq N}$,

$$|\Delta_N(\lambda)|^2 = \det \left(\left[\sum_{k=0}^{N-1} \lambda_i^k \lambda_j^k \right]_{1 \leq i, j \leq N} \right),$$

i.e. the determinant of a matrix containing polynomials in λ .

This form provides some progress, but it is particularly useful to transform these polynomials into the probabilistic Hermite polynomials¹ $He_k(x)$. These are monic polynomials of degree k , with $He_0(x) = 1$, $He_1(x) = x$, $He_2(x) = x^2 - 1$, for example; see [2] for a detailed discussion.

¹Sometimes called the Chebyshev-Hermite polynomials.

Importantly, these are monic polynomials and thus one can transform $V \rightarrow W$ with only column operations, where

$$W = \begin{bmatrix} He_0(\lambda_1) & He_1(\lambda_1) & \cdots & He_{N-1}(\lambda_1) \\ He_0(\lambda_2) & He_1(\lambda_2) & \cdots & He_{N-1}(\lambda_2) \\ \vdots & \vdots & \ddots & \vdots \\ He_0(\lambda_N) & He_1(\lambda_N) & \cdots & He_{N-1}(\lambda_N) \end{bmatrix}.$$

In the same sense as before, $WW^T = [\sum_{k=0}^{N-1} He_k(\lambda_i)He_k(\lambda_j)]_{1 \leq i,j \leq N}$ and therefore

$$|\Delta_N(\lambda)|^2 = \det \left(\left[\sum_{k=0}^{N-1} He_k(\lambda_i)He_k(\lambda_j) \right]_{1 \leq i,j \leq N} \right)$$

with the full chain of equality following $|\Delta_N(\lambda)|^2 = |\det(V)|^2 = \det(VV^T) = \det(WW^T)$.

Compare this expression with Ginibre's formula in (1). All that remains is to distribute the exponential into the determinant. Recall that if $A \rightarrow B$, where one row (or column) of A has been multiplied by a scalar c , then $\det(B) = c \det(A)$. Decomposing

$$e^{-|\lambda|^2/2} = \prod_{i=1}^N e^{-\lambda_i^2/4} \prod_{j=1}^N e^{-\lambda_j^2/4}$$

and distributing these terms, we arrive at the determinantal form of (1)

$$\rho_N(\lambda) = C_N \det \left(\left[\sum_{k=0}^{N-1} He_k(\lambda_i)e^{-\lambda_i^2/4}He_k(\lambda_j)e^{-\lambda_j^2/4} \right]_{1 \leq i,j \leq N} \right). \quad (3)$$

2. The Gaudin-Mehta formula. Hermite polynomials were chosen because they are orthogonal with respect to the measure $e^{-x^2/2}dx$. After normalising (3) with

$$\psi_k(x) = \frac{He_k(x)e^{-x^2/4}}{\sqrt{\sqrt{2\pi}k!}},$$

one can write ρ_N in terms of the kernel K_N with C_N now known,

$$K_N(x, y) := \sum_{k=0}^{N-1} \psi_k(x)\psi_k(y), \quad \rho_N(\lambda) = \frac{1}{N!} \det \left([K_N(\lambda_i, \lambda_j)]_{1 \leq i,j \leq N} \right).$$

This kernel is incredibly useful because of two main results: the trace identity

$$\int_{\mathbb{R}} K_N(x, x)dx = N$$

and the reproducing formula

$$\int_{\mathbb{R}} K_N(x, z)K_N(z, y)dz = K_N(x, y).$$

Together these allow for the *determinantal integration formula*, a process that reduces the dimension of the determinant when integrating out variables:

$$\int_{\mathbb{R}} \det[K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k+1} d\lambda_{k+1} = (N - k) \det[K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k}.$$

All details here are omitted (but will be tackled at a later date; for now, refer to [1]). By iterating this process (with Fubini-Tonelli) one can achieve the k -point correlation function ρ_k

$$\rho_k(\lambda_1, \dots, \lambda_k) = \det[K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k},$$

which yields the global eigenvalue distribution with $k = 1$. Thus the true probability density function of the expected empirical spectral distribution is given by

$$\frac{1}{N} K_N(\lambda_1, \lambda_1).$$

We will, later, continue analysis of K_N . In particular, obtaining the Christoffel-Darboux formula and then looking at the asymptotic behaviour as $N \rightarrow \infty$ is one way to derive Wigner's semicircular law (and possibly extend this to the circular law for Ginibre ensembles - matrices with all entries i.i.d complex standard normals).

It's currently understood that, in the setting of Dyson Brownian motion, one should instead use the orthonormal functions

$$\phi_k(x) = \frac{1}{c_k} e^{-NV(x)/2} \pi_k(x)$$

instead of ψ_k - the choice for $V(x) = x^2/2$. Here V is the potential of choice (e.g. $V(x) = qx^2 + gx^4$), c_k is the normalising constant, and $\pi_k(s)$ are the monic polynomials with respect to the measure $e^{-NV(s)} ds$ on $\mathbb{R}$. This is discussed well in [4].

References

- [1] Terence Tao. *Topics in random matrix theory*. Vol. 132. American Mathematical Society, 2023.
- [2] Gabor Szeg. *Orthogonal polynomials*. Vol. 23. American Mathematical Society, 1939.
- [3] Peter J Forrester. *Log-gases and random matrices*. Princeton University Press, 2010.
- [4] Xingjie Helen Li and Govind Menon. "Numerical solution of Dyson Brownian motion and a sampling scheme for invariant matrix ensembles". In: *Journal of Statistical Physics* 153.5 (2013), pp. 801–812.